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span: bounds-safe views for sequences of objects

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Changelog

Changes from R0

- Changed the name of the type being proposed from *array_view* to *span* following feedback from LEWG at the Kona meeting.
- Removed multidimensional aspects from the proposal. *span* is now always single-dimension and contiguous.
- Added details on potential interoperation with the multidimensional *view* type from P0009 [5].
- Removed functions to convert from *span<byte>* to *span<T>* as they are not compatible with type aliasing rules.
- Introduced dependency on P0257 [6] for definition of *byte* type, in order to support *span* as a method of accessing object representation.
- Added section containing proposed wording for inclusion in the standard.
- Simplified *span* interface based on reviewer feedback.

Changes from R1

- Added *difference_type* typedef to *span* to better support use in template functions.
- Removed *const_iterator begin const()* and *const_iterator end const ()* members of *span* based on LEWG feedback. For a view type like *span*, the constness of the view is immaterial to the constness of the element type, the iterator interface of *span* now reflects that.
- Removed the deletion of constructors that take rvalue-references based on LEWG feedback.
- Added support for construction from *const Container&*.

Changes from R2

- Wording cleanup: removed *const* on non-member functions and inappropriate *noexcept* specifiers. Improved wording to be clear that the *reverse_iterator* is not contiguous. Removed *constexpr* from *as_bytes()* and *as_writable_bytes()* as it would be illegal. Tidied up effects of *last()* overloads and of *array/std::array* constructors for cases when the array is empty.
- Added back *cbegin()* and *cend()* and *const_iterator* type based on LEWG feedback in Oulu.
- Improved colors.

Changes from R3

- Updated the wording to be differences against N4618.

Changes from R4

- Removed dependency on P0257 now that *byte* is part of the standard.
- Updated the wording to be differences against N4659.
- Added constructors from *unique_ptr*, *shared_ptr*.

- Removed unachievable *constexpr* from *as_bytes()* and *as_writeable_bytes()* functions.

Changes from R5

- Removed conversion constructors that took a *unique_ptr*/*shared_ptr* argument.
- Added *constexpr* qualifier to all iterator access functions on *span*.
- Removed *length()* and *length_bytes()* member functions from *span*. *length()* is considered unnecessary as *string_view* offers it if users are looking for *std::string* interface compatibility.
- Removed constructor from *span* that took a *nullptr_t* (as per request from LEWG). It does not add any value beyond the default constructor and may bind in unexpected ways for users.
- Removed move constructor and move assignment operator. They are unnecessary as this is designed to be a copy-only type.
- Removed redundant “Effects” clause from descriptions of copy constructor and assignment operator in proposed wording.
- Simplified many member functions descriptions down to an “effects equivalent to” form in proposed wording.
- Corrected typo in description of *as_writeable_bytes()* function.
- Added covering statement to synopsis that marks all member functions as having constant time complexity and removed individual time complexity clauses to proposed wording.
- Added (accidentally-) missing description for *cbegin()/cend()/crbegin()/crend()* to proposed wording.
- Removed unnecessary *std::* qualification from *remove_cv_t()* call in proposed wording.
- Corrected definitions of comparison operations to take arguments by-value rather than by-reference to reflect the design of *span* as a copy-only type.
- Removed incorrect italicization of *byte* in proposed wording.

Changes from R6

- Modified wording of *subspan<Offset, Count>()* to reflect the preferred design: that a fixed-size *span* type is returned wherever possible, and a dynamic-size one is returned only as a fallback. So that, as an example, *span<int, 42>.subspan<2>()* will return a *span<int, 40>*.
- Modified wording of “from-container” constructor to reflect a simpler design, as encouraged by LWG/LEWG. Now the container requirements are just that *std::size()* and *std::data()* work for the container, and that the return of *std::data()* is convertible to the pointer-type of the *span*.
- Removed the constructors that took a *std::array*, as these can be better served via the from-container constructor now.
- Ensure the from-container constructors are consistently declared with *constexpr* in the document
- Described the behavior of *span::begin()* when the *span* is empty.
- Added updates to wording in Iterators section to ensure that free functions *begin()*, *end()*, *empty()*, *data()*, *size()* are also specialized for *span*.
- Fixed typos. (Changed by STL, and below.)
- Added deduction guides.
- Marked *dynamic_extent* as *inline*.
- Fixed section numbers; this has always been proposed for the Containers clause, now 26.
- Removed comment duplicating *[views.span]/1*.

- Constrained default constructor properly: only zero-fixed-extent and dynamic-extent spans are default constructible.
- Fixed Throws element of the Container constructors.
- Reordered Container constructor `is_same_v` check, and changed `remove_cvref_t` to `remove_cv_t`.
- Fixed Container constraint to prevent `span<Derived>` converting to `span<Base>`.
- Constrained built-in array constructor, instead of ill-formed enforcement.
- Removed Throws Nothing from built-in array constructor, which is already noexcept.
- Renamed `as_writeable_bytes` to `as_writable_bytes`.
- Replaced `distance(firstElem, lastElem)` with `lastElem - firstElem`, as they are pointers.
- Changed one occurrence of `cont.size()` to `size(cont)`.
- Used “valid range” to simplify requirements.
- Simplified “If `ptr` is null or `count` is 0” to “If `count` is 0”.
- Added wording to `cbegin()` for consistency with `begin()`.
- In the header synopsis, changed `as_bytes` and `as_writable_bytes` to use `byte` instead of `char`.
- In the class synopsis, removed declarations of `as_bytes` and `as_writable_bytes` (which were missing noexcept).
- The header synopsis now declares the heterogeneous comparisons, which are not repeated in the class synopsis.
- Added missing template arguments in `[span.sub]` (copied from return types).
- Changed `subspan()` to return `span<ElementType, Count != dynamic_extent ? Count : (Extent != dynamic_extent ? Extent - Offset : Extent)>`. This changed `Extent - Offset - 1` to `Extent - Offset`. If the user asks for a subspan with dynamic `Count`, but we have a fixed `Extent`, then we will return `Extent - Offset` elements. For example, `Extent == 5, Offset == 0` asks for a full subspan; we return `5 - 0 == 5` elements. (Confirmed by Neil)
- Filled in `subspan()`’s returned span with `(data() + Offset, Count != dynamic_extent ? Count : (Extent != dynamic_extent ? Extent - Offset : size() - Offset))`. (Confirmed by Neil)
- Changed the first part of `subspan()`’s requirement to `(Offset >= 0 && Offset <= size())`, always permitting `Offset == size()`. (Confirmed by Neil)
- Also changed `subspan(offset, count)`’s first requirement to `(offset >= 0 && offset <= size())`. (Confirmed by Neil)
- Changed `"sizeof(ElementType) * Extent"` to `"static_cast<ptrdiff_t>(sizeof(ElementType)) * Extent"` in `as_bytes` and `as_writable_bytes` to avoid forbidden narrowing.
- Restored `std::array` constructors (whose declarations were still present), combined their specification with the built-in array constructor.
- Changed `array<remove_const_t<element_type>, N>` to `array<value_type, N>`. `array`’s template parameter shouldn’t be cv-qualified, especially volatile-qualified.
- Marked the built-in array and `std::array` constructors as noexcept in declaration and definition.
- Dropped “The `reverse_iterator` type is a random access iterator.” as it is completely redundant with the specification that uses `std::reverse_iterator` of the (random-access) iterator type.
- Reworked array constructor constraints (`span<Object>` shouldn’t be constructible from `const array<Object, N>`).

- Reworked Container constructor constraints: now it avoids competing with built-in arrays, `std::arrays`, and any `std::spans` (including converting).
- Fixed `crbegin/crend`'s definitions to use `const_reverse_iterator`.
- In `operator<=()`, changed `return !(l > r);` to `return !(r < l);` to flatten the callstack.
- For clarity, changed `as_bytes()` and `as_writable_bytes()` to call `s.size_bytes()`.
- Changed to returning spans with `{}` for less verbosity, following `as_bytes/as_writable_bytes`.
- Changed `<class ElementL, ptrdiff_t ExtentL, class ElementR, ptrdiff_t ExtentR>` to `<class T, ptrdiff_t X, class U, ptrdiff_t Y>` in order to reduce repetitive verbosity.
- Overhauled `span`'s converting constructor: new constraint supersedes `Requires`, constructor is `noexcept`.
- Added wording to update Annex C, which also forgot `<charconv>`. (That was C++17 P0067R5 Elementary String Conversions, updated by P0682R1 Repairing Elementary String Conversions as a Defect Report in Toronto, so the header is part of C++17 and not just C++20.)
- Added `<span>` to `[iterator.range]`.
- Added `<string_view>` to `[iterator.container]`. It has all of `size()`, `empty()`, and `data()`.
- Changed `"constexpr static"` to `"static constexpr"` which is consistently used in the Standard.

Introduction

This paper presents a design for a fundamental vocabulary type *span*.

The *span* type is an abstraction that provides a view over a contiguous sequence of objects, the storage of which is owned by some other object. The design for *span* presented here provides bounds-safety guarantees through a combination of compile-time and (configurable) run-time constraints.

The design of the *span* type discussed in this paper is related to the *span* previously proposed in N3851 [1] and also draws on ideas in the *array_ref* and *string_ref* classes proposed in N3334 [2]. *span* is closely related to the generalized, multidimensional memory-access abstraction *array_ref* described in P0009 [5]. The *span* proposed here is sufficiently compatible with *array_ref* that interoperability between the two types would be simple and well-defined.

While *array_ref* is proposed by P0009 [5] as a generalized and highly configurable view type that can address needs for specialized domains such as scientific computing, *span* is proposed as a simple solution to the common need for a single-dimensional view over contiguous storage.

Motivation and Scope

The evolution of the standard library has demonstrated that it is possible to design and implement abstractions in Standard C++ that improve the reliability of C++ programs without sacrificing either performance or portability. This proposal identifies a new “vocabulary type” for inclusion in the standard library that enables both high performance and bounds-safe access to contiguous sequences of elements. This type would also improve modularity, composability, and reuse by decoupling accesses to array data from the specific container types used to store that data.

These characteristics lead to higher quality programs. Some of the bounds and type safety constraints of *span* directly support “correct-by-construction” programming methodology – where errors simply do not compile. One of the major advantages of *span* over the common idiom of a “pointer plus length” pair of parameters is that it provides clearer semantics hints to analysis tools looking to help detect and prevent defects early in a software development cycle.

Impact on the Standard

This proposal is a pure library extension. It does not require any changes to standard classes, functions, or headers.

However – if adopted – it may be useful to overload some standard library functions for this new type (an example would be *copy()*).

span has been implemented in standard C++ (C++11) and is being successfully used within a commercial static analysis tool for C++ code as well as commercial office productivity software. An open source, reference implementation is available at <https://github.com/Microsoft/GSL> [3].

Design Decisions

View not container

span is simply a view over another object's contiguous storage – but unlike *array* or *vector* it does not “own” the elements that are accessible through its interface. An important observation arises from this: *span* never performs any free store allocations.

While *span* is a view, it is not an iterator. You cannot perform increment or decrement operations on it, nor dereference it.

No configurable view properties

In the related *array_ref* type described in P0009 [5], properties are used to control policies such as memory layout (column-major, row-major) and location (on heterogeneous memory architectures) for specific specializations of *array_ref*. *span* does not require properties as it is always a simple view over contiguous storage. Its memory layout and access characteristics are equivalent to those of a built-in array. This difference should not prevent conversions between *array_ref* and *span* instances, it merely constrains that they could only be available in cases where *array_ref* properties are compatible with the characteristics of *span*.

View length and measurement

The general usage protocol of the *span* class template supports both static-size (fixed at compile time) and dynamic-size (provided at runtime) views. The *Extent* template parameter to *span* is used to provide the extent of the *span*.

```
constexpr ptrdiff_t dynamic_extent = -1;
```

The default value for *Extent* is *dynamic_extent*: a unique value outside the normal range of lengths (0 to *PTRDIFF_MAX* inclusive) reserved to indicate that the length of the sequence is only known at runtime and must be stored within the *span*. A dynamic-size *span* is, conceptually, just a pointer and size field (this is not an implementation requirement, however).

```
int* somePointer = new int[someLength];

// Declaring a dynamic-size span
// s will have a dynamic-size specified by someLength at construction
span<int> s { somePointer, someLength };
```

The type used for measuring and indexing into *span* is *ptrdiff_t*. Using a signed index type helps avoid common mistakes that come from implicit signed to unsigned integer conversions when users employ integer literals (which are nearly always signed). The use of *ptrdiff_t* is natural as it is the type used for pointer arithmetic and array indexing – two operations that *span* explicitly aims to replace but that an implementation of *span* would likely rely upon.

A fixed-size *span* provides a value for *Extent* that is between 0 and *PTRDIFF_MAX* (inclusive). A fixed-size *span* requires no storage size overhead beyond a single pointer – using the type system to carry the fixed-

length information. This allows *span* to be an extremely efficient type to use for access to fixed-length buffers.

```
int arr[10];

// deduction of size from arrays means that span size is always correct
span<int, 10> s2 { arr }; // fixed-size span of 10 ints
span<int, 20> s3 { arr }; // error: will fail compilation
span<int> s4 { arr }; // dynamic-size span of 10 ints
```

Value Type Semantics

span is designed as a value type – it is expected to be cheap to construct, copy, move, and use. Users are encouraged to use it as a pass-by-value parameter type wherever they would have passed a pointer by value or a container type by reference, such as *array* or *vector*.

Conceptually, *span* is simply a pointer to some storage and a count of the elements accessible via that pointer. Those two values within a span can only be set via construction or assignment (i.e. all member functions other than constructors and assignment operators are *const*). This property makes it easy for users to reason about the values of a span through the course of a function body.

These value type characteristics also help provide compiler implementations with considerable scope for optimizing the use of *span* within programs. For example, *span* has a trivial destructor, so common ABI conventions allow it to be passed in registers.

Range-checking and bounds-safety

All accesses to the data encapsulated by a span are conceptually range-checked to ensure they remain within the bounds of the *span*. What actually happens as the result of a failure to meet *span*'s bounds-safety constraints at runtime is undefined behavior. However, it should be considered effectively fatal to a program's ability to continue reliable execution. This is a critical aspect of *span*'s design, and allows users to rely on the guarantee that as long as a sequence is accessed via a correctly initialized *span*, then its bounds cannot be overrun.

As an example, in the current reference implementation, violating a range-check results by default in a call to *terminate()* but can also be configured via build-time mechanisms to continue execution (albeit with undefined behavior from that point on).

Conversion between fixed-size and dynamic-size *span* objects is allowed, but with strict constraints that ensure bounds-safety is always preserved. At least two of these cases can be checked statically by leveraging the type system. In each case, the following rules assume the element types of the *span* objects are compatible for assignment.

1. A fixed-size *span* may be constructed or assigned from another fixed-size span of equal length.
2. A dynamic-size *span* may always be constructed or assigned from a fixed-size *span*.
3. A fixed-size *span* may always be constructed or assigned from a dynamic-size *span*. Undefined behavior will result if the construction or assignment is not bounds-safe. In the reference

implementation, for example, this is achieved via a runtime check that results in *terminate()* on failure.

Element types and conversions

span must be configured with its element type via the template parameter *ValueType*, which is required to be a complete object type that is not an abstract class type. *span* supports either read-only or mutable access to the sequence it encapsulates. To access read-only data, the user can declare a *span<const T>*, and access to mutable data would use a *span<T>*.

Construction or assignment between *span* objects with different element types is allowed whenever it can be determined statically that the element types are exactly storage-size equivalent (so there is no difference in the extent of memory being accessed), and that the types can legally be aliased.

As a result of these rules, it is always possible to convert from a *span<T>* to a *span<const T>*. It is not allowed to convert in the opposite direction, from *span<const T>* to *span<T>*. This property is extremely convenient for calling functions that take *span* parameters.

Element access and iteration

span's interface for accessing elements is largely similar to that of *array*. It overloads *operator[]* for element access, and offers random access iterators, making it adoptable with a minimum of source changes in code that previously used an array, an *array* object, or a pointer to access more than one object. *span* also overloads *operator()* for element access, to provide compatibility with code written to operate against *view*.

span provides random-access iterators over its data, comparable to *vector* and *array*. All accesses to elements made through these iterators are range-checked (subject to configuration as previously described), just as if they had been performed via the subscript operator on *span*. There is no difference in the mutability of the iterators returned from a *const* or non-*const* *span* as the constness of the element type is already determined when the *span* is created. As is appropriate for a view, whether the *span* itself is *const* does not affect the element type, and this is reflected in the simplicity of the iterator model.

```
// [span.elem], span element access
constexpr reference operator[](index_type idx) const;
constexpr reference operator()(index_type idx) const;
constexpr pointer data() const noexcept;

// [span.iter], span iterator support
constexpr iterator begin() const noexcept;
constexpr iterator end() const noexcept;

constexpr const_iterator cbegin() const noexcept;
constexpr const_iterator cend() const noexcept;

constexpr reverse_iterator rbegin() const noexcept;
constexpr reverse_iterator rend() const noexcept;

constexpr const_reverse_iterator crbegin() const noexcept;
constexpr const_reverse_iterator crend() const noexcept;
```

Construction

The *span* class is expected to become a frequently used vocabulary type in function interfaces (as a safer replacement of “(pointer, length)” idioms), as it specifies a minimal set of requirements for safely accessing a sequence of objects and decouples a function that needs to access a sequence from the details of the storage that holds such elements.

To simplify use of *span* as a simple parameter, *span* offers a number of constructors for common container types that store contiguous sequences of elements. A summarized extract from the specification illustrates this:

```
// [span.cons], span constructors, copy, assignment, and destructor
constexpr span();
constexpr span(pointer ptr, index_type count);
constexpr span(pointer firstElem, pointer lastElem);
template <size_t N>
    constexpr span(element_type (&arr)[N]);
template <size_t N>
    constexpr span(array<remove_const_t<element_type>, N>& arr);
template <size_t N>
    constexpr span(const array<remove_const_t<element_type>, N>& arr);
template <class Container>
    constexpr span(Container& cont);
template <class Container>
    constexpr span(const Container& cont);
constexpr span(const span& other) noexcept = default;
template <class OtherElementType, ptrdiff_t OtherExtent>
    constexpr span(const span<OtherElementType, OtherExtent>& other);
```

It is allowed to construct a span from the null pointer, and this creates an object with *.size() == 0*. Any attempt to construct a span with a null pointer value and a non-zero length is considered a range-check error.

Byte representations and conversions

A span of any element type that is a standard-layout type can be converted to a *span<const byte>* or a *span<byte>* via the free functions *as_bytes()* and *as_writable_bytes()* respectively. These operations are considered useful for systems programming where byte-oriented access for serialization and data transmission is essential.

```
// [span.objectrep], views of object representation
template <class ElementType, ptrdiff_t Extent>
    span<const byte, ((Extent == dynamic_extent) ? dynamic_extent :
    (sizeof(ElementType) * Extent))> as_bytes(span<ElementType, Extent> s)
    noexcept;

template <class ElementType, ptrdiff_t Extent>
```

```
span<byte,      ((Extent == dynamic_extent) ? dynamic_extent :
(sizeof(ElementType) * Extent))> as_writable_bytes(span<ElementType, Extent>
) noexcept;
```

These byte-representation conversions still preserve const-correctness, however. It is not possible to convert from a *span<const T>* to a *span<byte>* (through SFINAE overload restriction).

Comparisons

span supports all the same comparison operations as a sequential standard library container: element-wise comparison and a total ordering by lexicographical comparison. This helps make it an effective replacement for existing uses of sequential contiguous container types like *array* or *vector*.

```
// [span.comparison], span comparison operators
template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator==(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);

template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator!=(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);

template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator<(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);

template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator<=(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);

template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator>(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);

template <class ElementL, ptrdiff_t ExtentL,
         class ElementR, ptrdiff_t ExtentR>
constexpr bool operator>=(span<ElementL, ExtentL> l, span<ElementR, ExtentR>
r);
```

Regardless of whether they contain a valid pointer or null pointer, zero-length *spans* are all considered equal. This is considered a useful property when writing library code. If users wish to distinguish between a zero-length *span* with a valid pointer value and a *span* containing the null pointer, then they can do so by calling the *data()* member function and examining the pointer value directly.

Creating sub-spans

span offers convenient member functions for generating a new *span* that is a reduced view over its sequence. In each case, the newly constructed *span* is returned by value from the member function. As the design requires bounds-safety, these member functions are guaranteed to either succeed and return a valid *span*, or fail with undefined behavior (e.g. calling *terminate()*) if the parameters were not within range.

```
// [span.sub], span subviews
constexpr span<element_type, dynamic_extent> first(index_type count) const;
constexpr span<element_type, dynamic_extent> last(index_type count) const;
constexpr span<element_type, dynamic_extent> subspan(index_type offset,
index_type count = dynamic_extent) const;
```

first() returns a new *span* that is limited to the first N elements of the original sequence. Conversely, *last()* returns a new *span* that is limited to the last N elements of the original sequence. *subspan()* allows an arbitrary sub-range within the sequence to be selected and returned as a new *span*.

All three member functions are overloaded in forms that accept their parameters as template parameters, rather than function parameters. These overloads are helpful for creating fixed-size *span* objects from an original input *span*, whether fixed- or dynamic-size.

```
template <ptrdiff_t Count>
    constexpr span<element_type, Count> first() const;
template <ptrdiff_t Count>
    constexpr span<element_type, Count> last() const;
template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
    constexpr span<element_type, /* see wording */> subspan() const;
```

Multidimensional *span*

span as presented here only supports a single-dimension view of a sequence. This covers the most common usage of contiguous sequences in C++. *span* has convenience (such as iterators, *first()*, *last()*, and *subspan()*) and default behaviors that make most sense in a single-dimension.

Adding support for multidimensional and noncontiguous (strided) views of data is deferred to a separate type not described here. One such candidate would be the more general *array_ref* facility described in P0009 [5]. The interface of *span* is sufficiently compatible with that of *array_ref*, that users should not feel any significant discontinuity between the two. In fact, it is entirely possible to implement a *span* using *array_ref*.

Proposed Wording Changes

The following proposed wording changes against the working draft of the standard are relative to N4659 [6].

20.5.1.2 Headers [headers]

2 The C++ standard library provides the *C++ library headers*, as shown in Table 16.

Table 16 – C++ library headers

<algorithm>	<future>	<numeric>	<string_view>
<any>	<initializer_list>	<optional>	<stringstream>
<array>	<iomanip>	<ostream>	<system_error>
<atomic>	<ios>	<queue>	<thread>
<bitset>	<iosfwd>	<random>	<tuple>
<chrono>	<iostream>	<ratio>	<type_traits>
<codecvt>	<istream>	<regex>	<typeindex>
<complex>	<iterator>	<scoped_allocator>	<typeinfo>
<condition_variable>	<limits>	<set>	<unordered_map>
<deque>	<list>	<shared_mutex>	<unordered_set>
<exception>	<locale>		<utility>
<execution>	<map>	<sstream>	<valarray>
<filesystem>	<memory>	<stack>	<variant>
<forward_list>	<memory_resources>	<stdexcept>	<vector>
<fstream>	<mutex>	<streambuf>	
<functional>	<new>	<string>	

26 Containers library [containers]

26.1 General [containers.general]

2 The following subclauses describe container requirements, and components for sequence containers, associative containers, and views as summarized in Table 82.

Table 82 – Containers library summary

Subclause	Header(s)
26.2 Requirements	
26.3 Sequence containers	<array> <deque> <forward_list> <list> <vector>
26.4 Associative containers	<map> <set>
26.5 Unordered associative containers	<unordered_map> <unordered_set>
26.6 Container adaptors	<queue> <stack>
<u>26.7 Views</u>	<u></u>

27 Iterators library [iterators]

27.7 Range access [iterator.range]

1 In addition to being available via inclusion of the <iterator> header, the function templates in 27.7 are available when any of the following headers are included: <array>, <deque>, <forward_list>, <list>, <map>, <regex>, <set>, , <string>, <string_view>, <unordered_map>, <unordered_set>, and <vector>.

27.8 Container and view access [iterator.container]

1 In addition to being available via inclusion of the `<iterator>` header, the function templates in 27.8 are available when any of the following headers are included: `<array>`, `<deque>`, `<forward_list>`, `<list>`, `<map>`, `<regex>`, `<set>`, `<span>`, `<string>`, `<string_view>`, `<unordered_map>`, `<unordered_set>`, and `<vector>`.

26.7 Views [views]

26.7.1 General [views.general]

1 The header `<span>` defines the view `span`. A `span` is a view over a contiguous sequence of objects, the storage of which is owned by some other object.

Header `<span>` synopsis

```
namespace std {

// constants
inline constexpr ptrdiff_t dynamic_extent = -1;

// [views.span], class template span
template <class ElementType, ptrdiff_t Extent = dynamic_extent>
class span;

// [span.comparison], span comparison operators
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator==(span<T, X> l, span<U, Y> r);

template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator!=(span<T, X> l, span<U, Y> r);

template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator<(span<T, X> l, span<U, Y> r);

template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator<=(span<T, X> l, span<U, Y> r);

template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator>(span<T, X> l, span<U, Y> r);

template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator>=(span<T, X> l, span<U, Y> r);

// [span.objectrep], views of object representation
template <class ElementType, ptrdiff_t Extent>
span<const byte, ((Extent == dynamic_extent) ? dynamic_extent :
(static_cast<ptrdiff_t>(sizeof(ElementType)) * Extent))>
as_bytes(span<ElementType, Extent> s) noexcept;

template <class ElementType, ptrdiff_t Extent>
```

```

    span<byte,      ((Extent == dynamic_extent) ? dynamic_extent :
    (static_cast<ptrdiff_t>(sizeof(ElementType)) * Extent))>
    as_writable_bytes(span<ElementType, Extent> s) noexcept;
} // namespace std

```

26.7.2 Class template `span` [views.span]

- 1 A `span` is a view over a contiguous sequence of objects, the storage of which is owned by some other object.
- 2 `ElementType` is required to be a complete object type that is not an abstract class type.
- 3 If `Extent < dynamic_extent`, the program is ill-formed.
- 4 The iterator type for `span` is a random access iterator and contiguous iterator.
- 5 All member functions of `span` have constant time complexity.

```

namespace std {
template <class ElementType, ptrdiff_t Extent = dynamic_extent>
class span {
public:
    // constants and types
    using element_type = ElementType;
    using value_type = remove_cv_t<ElementType>;
    using index_type = ptrdiff_t;
    using difference_type = ptrdiff_t;
    using pointer = element_type*;
    using reference = element_type&;
    using iterator = /* implementation-defined */;
    using const_iterator = /* implementation-defined */;
    using reverse_iterator = std::reverse_iterator<iterator>;
    using const_reverse_iterator = std::reverse_iterator<const_iterator>;

    static constexpr index_type extent = Extent;

    // [span.cons], span constructors, copy, assignment, and destructor
    constexpr span() noexcept;
    constexpr span(pointer ptr, index_type count);
    constexpr span(pointer firstElem, pointer lastElem);
    template <size_t N>
        constexpr span(element_type (&arr)[N]) noexcept;
    template <size_t N>
        constexpr span(array<value_type, N>& arr) noexcept;
    template <size_t N>
        constexpr span(const array<value_type, N>& arr) noexcept;
    template <class Container>
        constexpr span(Container& cont);
    template <class Container>

```

```

    constexpr span(const Container& cont);
    constexpr span(const span& other) noexcept = default;
    template <class OtherElementType, ptrdiff_t OtherExtent>
        constexpr span(const span<OtherElementType, OtherExtent>& s) noexcept;
    ~span() noexcept = default;
    constexpr span& operator=(const span& other) noexcept = default;

    // [span.sub], span subviews
    template <ptrdiff_t Count>
        constexpr span<element_type, Count> first() const;
    template <ptrdiff_t Count>
        constexpr span<element_type, Count> last() const;
    template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
        constexpr span<element_type, /* see below */> subspan() const;
    constexpr span<element_type, dynamic_extent> first(index_type count)
const;
    constexpr span<element_type, dynamic_extent> last(index_type count)
const;
    constexpr span<element_type, dynamic_extent> subspan(index_type offset,
index_type count = dynamic_extent) const;

    // [span.obs], span observers
    constexpr index_type size() const noexcept;
    constexpr index_type size_bytes() const noexcept;
    constexpr bool empty() const noexcept;

    // [span.elem], span element access
    constexpr reference operator[](index_type idx) const;
    constexpr reference operator()(index_type idx) const;
    constexpr pointer data() const noexcept;

    // [span.iterators], span iterator support
    constexpr iterator begin() const noexcept;
    constexpr iterator end() const noexcept;

    constexpr const_iterator cbegin() const noexcept;
    constexpr const_iterator cend() const noexcept;

    constexpr reverse_iterator rbegin() const noexcept;
    constexpr reverse_iterator rend() const noexcept;

    constexpr const_reverse_iterator crbegin() const noexcept;
    constexpr const_reverse_iterator crend() const noexcept;

private:
    pointer data_; // exposition only
    index_type size_; // exposition only
};

template<class T, size_t N>
    span(T (&)[N]) -> span<T, N>;

```



```

template<class T, size_t N>
    span(array<T, N>&) -> span<T, N>;

template<class T, size_t N>
    span(const array<T, N>&) -> span<const T, N>;

template<class Container>
    span(Container&) -> span<typename Container::value_type>;

template<class Container>
    span(const Container&) -> span<const typename Container::value_type>;

} // namespace std

```

26.7.2.1 span constructors, copy, assignment, and destructor [span.cons]

```
constexpr span() noexcept;
```

Remarks: This constructor shall not participate in overload resolution unless `Extent <= 0` is true.

Effects: Constructs an empty `span`.

Postconditions: `size() == 0 && data() == nullptr`

```
constexpr span(pointer ptr, index_type count);
```

Requires: `[ptr, ptr + count)` shall be a valid range. If `extent` is not equal to `dynamic_extent`, then `count` shall be equal to `extent`.

Effects: Constructs a `span` that is a view over the range `[ptr, ptr + count)`. If `count` is 0 then an empty `span` is constructed.

Postconditions: `size() == count && data() == ptr`

Throws: Nothing.

```
constexpr span(pointer firstElem, pointer lastElem);
```

Requires: `[firstElem, lastElem)` shall be a valid range. If `extent` is not equal to `dynamic_extent`, then `lastElem - firstElem` shall be equal to `extent`.

Effects: Constructs a `span` that is a view over the range `[firstElem, lastElem)`. If `lastElem - firstElem == 0` then an empty `span` is constructed.

Postconditions: `size() == lastElem - firstElem && data() == firstElem`

Throws: Nothing.

```
template <size_t N>
    constexpr span(element_type (&arr) [N]) noexcept;
template <size_t N>
    constexpr span(array<value_type, N>& arr) noexcept;
template <size_t N>
    constexpr span(const array<value_type, N>& arr) noexcept;
```

Remarks: These constructors shall not participate in overload resolution unless:

- `extent == dynamic_extent || N == extent` is true, and
- `remove_pointer_t<decltype(data(arr))>(*) []` is convertible to `ElementType(*) []`.

Effects: Constructs a `span` that is a view over the supplied array.

Postconditions: `size() == N && data() == data(arr)`

```
template <class Container>
    constexpr span(Container& cont);
template <class Container>
    constexpr span(const Container& cont);
```

Remarks: These constructors shall not participate in overload resolution unless:

- `Container` is not a specialization of `span`,
- `Container` is not a specialization of `array`,
- `is_array_v<Container>` is false,
- `data(cont)` and `size(cont)` are both well-formed, and
- `remove_pointer_t<decltype(data(cont))>(*) []` is convertible to `ElementType(*) []`.

Requires: `[data(cont), data(cont) + size(cont))` shall be a valid range. If `extent` is not equal to `dynamic_extent`, then `size(cont)` shall be equal to `extent`.

Effects: Constructs a `span` that is a view over the range `[data(cont), data(cont) + size(cont))`.

Postconditions: `size() == size(cont) && data() == data(cont)`

Throws: What and when `data(cont)` and `size(cont)` throw.

```
constexpr span(const span& other) noexcept = default;
```

Postconditions: `other.size() == size() && other.data() == data()`

```
template <class OtherElementType, ptrdiff_t OtherExtent>
constexpr span(const span<OtherElementType, OtherExtent>& s) noexcept;
```

Remarks: This constructor shall not participate in overload resolution unless:

- `Extent == dynamic_extent || Extent == OtherExtent` is true,
- `OtherElementType(*)[]` is convertible to `ElementType(*)[]`.

Effects: Constructs a span that is a view over the range `[s.data(), s.data() + s.size())`.

Postconditions: `size() == s.size() && data() == s.data()`

```
constexpr span& operator=(const span& other) noexcept = default;
```

Postconditions: `size() == other.size() && data() == other.data()`

26.7.2.2 span subviews [span.sub]

```
template <ptrdiff_t Count>
constexpr span<element_type, Count> first() const;
```

Requires: `Count >= 0 && Count <= size()`

Effects: Equivalent to: `return { data(), Count };`

```
template <ptrdiff_t Count>
constexpr span<element_type, Count> last() const;
```

Requires: `Count >= 0 && Count <= size()`

Effects: Equivalent to: `return { data() + (size() - Count), Count };`

```
template <ptrdiff_t Offset, ptrdiff_t Count = dynamic_extent>
constexpr span<element_type, /* see below */> subspan() const;
```

Requires: `(Offset >= 0 && Offset <= size()) && (Count == dynamic_extent || Count >= 0 && Offset + Count <= size())`

Effects: Equivalent to: `return span<ElementType, Count != dynamic_extent ? Count : (Extent != dynamic_extent ? Extent - Offset : dynamic_extent)>(data() + Offset, Count != dynamic_extent ? Count : (Extent != dynamic_extent ? Extent - Offset : size() - Offset));`

```
constexpr span<element_type, dynamic_extent> first(index_type count)
const;
```

Requires: `count >= 0 && count <= size()`

Effects: Equivalent to: `return { data(), count };`

```
constexpr span<element_type, dynamic_extent> last(index_type count)
const;
```

Requires: `count >= 0 && count <= size()`

Effects: Equivalent to: `return { data() + (size() - count), count };`

```
constexpr span<element_type, dynamic_extent> subspan(index_type offset,
index_type count = dynamic_extent) const;
```

Requires: `(offset >= 0 && offset <= size()) && (count == dynamic_extent || count >= 0 && offset + count <= size())`

Effects: Equivalent to: `return { data() + offset, count == dynamic_extent ? size() - offset : count };`

26.7.2.2 span observers [span.obs]

```
constexpr index_type size() const noexcept;
```

Effects: Equivalent to: `return size_;`

```
constexpr index_type size_bytes() const noexcept;
```

Effects: Equivalent to: `return size() * sizeof(element_type);`

```
constexpr bool empty() const noexcept;
```

Effects: Equivalent to: `return size() == 0;`

26.7.2.3 span element access [span.elem]

```
constexpr reference operator[](index_type idx) const;  
constexpr reference operator()(index_type idx) const;
```

Requires: `idx >= 0 && idx < size()`

Effects: Equivalent to: `return *(data() + idx);`

```
constexpr pointer data() const noexcept;
```

Effects: Equivalent to: `return data_;`

26.7.2.4 span iterator support [span.iterators]

```
constexpr iterator begin() const noexcept;
```

Returns: An iterator referring to the first element in the span. If `empty()` is `true`, then it returns the same value as `end()`.

```
constexpr iterator end() const noexcept;
```

Returns: An iterator which is the past-the-end value.

```
constexpr reverse_iterator rbegin() const noexcept;
```

Effects: Equivalent to `return reverse_iterator(end());`

```
constexpr reverse_iterator rend() const noexcept;
```

Returns: Equivalent to: `return reverse_iterator(begin());`

```
constexpr const_iterator cbegin() const noexcept;
```

Returns: A constant iterator referring to the first element in the `span`. If `empty()` is `true`, then it returns the same value as `cend()`.

```
constexpr const_iterator cend() const noexcept;
```

Returns: A constant iterator which is the past-the-end value.

```
constexpr const_reverse_iterator crbegin() const noexcept;
```

Effects: Equivalent to `return const_reverse_iterator(cend())`.

```
constexpr const_reverse_iterator crend() const noexcept;
```

Returns: Equivalent to: `return const_reverse_iterator(cbegint());`

26.7.2.5 span comparison operators [span.comparison]

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>  
constexpr bool operator==(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return equal(l.begin(), l.end(), r.begin(), r.end());`

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>  
constexpr bool operator!=(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return !(l == r);`

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>  
constexpr bool operator<(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return lexicographical_compare(l.begin(), l.end(), r.begin(), r.end());`

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator<=(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return !(r < l);`

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator>(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return (r < l);`

```
template <class T, ptrdiff_t X, class U, ptrdiff_t Y>
constexpr bool operator>=(span<T, X> l, span<U, Y> r);
```

Effects: Equivalent to: `return !(l < r);`

26.7.2.6 views of object representation [span.objectrep]

```
template <class ElementType, ptrdiff_t Extent>
span<const byte, ((Extent == dynamic_extent) ? dynamic_extent :
(static_cast<ptrdiff_t>(sizeof(ElementType)) * Extent))>
as_bytes(span<ElementType, Extent> s) noexcept;
```

Effects: Equivalent to: `return { reinterpret_cast<const byte*>(s.data()), s.size_bytes() };`

```
template <class ElementType, ptrdiff_t Extent>
span<byte, ((Extent == dynamic_extent) ? dynamic_extent :
(static_cast<ptrdiff_t>(sizeof(ElementType)) * Extent))>
as_writable_bytes(span<ElementType, Extent> s) noexcept;
```

Remarks: This function shall not participate in overload resolution unless `is_const_v<ElementType>` is false.

Effects: Equivalent to: `return { reinterpret_cast<byte*>(s.data()), s.size_bytes() };`

C.4.8 Clause 20: library introduction [diff.cpp14.library]

1 Affected subclause: 20.5.1.2

Change: New headers.

Rationale: New functionality.

Effect on original feature: The following C++ headers are new: `<any>`, `<charconv>`, `<execution>`, `<filesystem>`, `<memory_resource>`, `<optional>`, `<string_view>`, and `<variant>`. Valid C++ 2014 code that `#includes` headers with these names may be invalid in this International Standard.

C.5.4 Clause 20: library introduction [diff.cpp17.library]

1 Affected subclause: 20.5.1.2

Change: New headers.

Rationale: New functionality.

Effect on original feature: The following C++ headers are new: `<compare>`, `<span>`, and `<syncstream>`. Valid C++ 2017 code that `#includes` headers with these names may be invalid in this International Standard.

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The authors of P0009 were invaluable in discussing how *span* and *array_ref* can be compatible and by doing so support a programming model that is safe and consistent as users move between a single dimension and multiple dimensions.

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